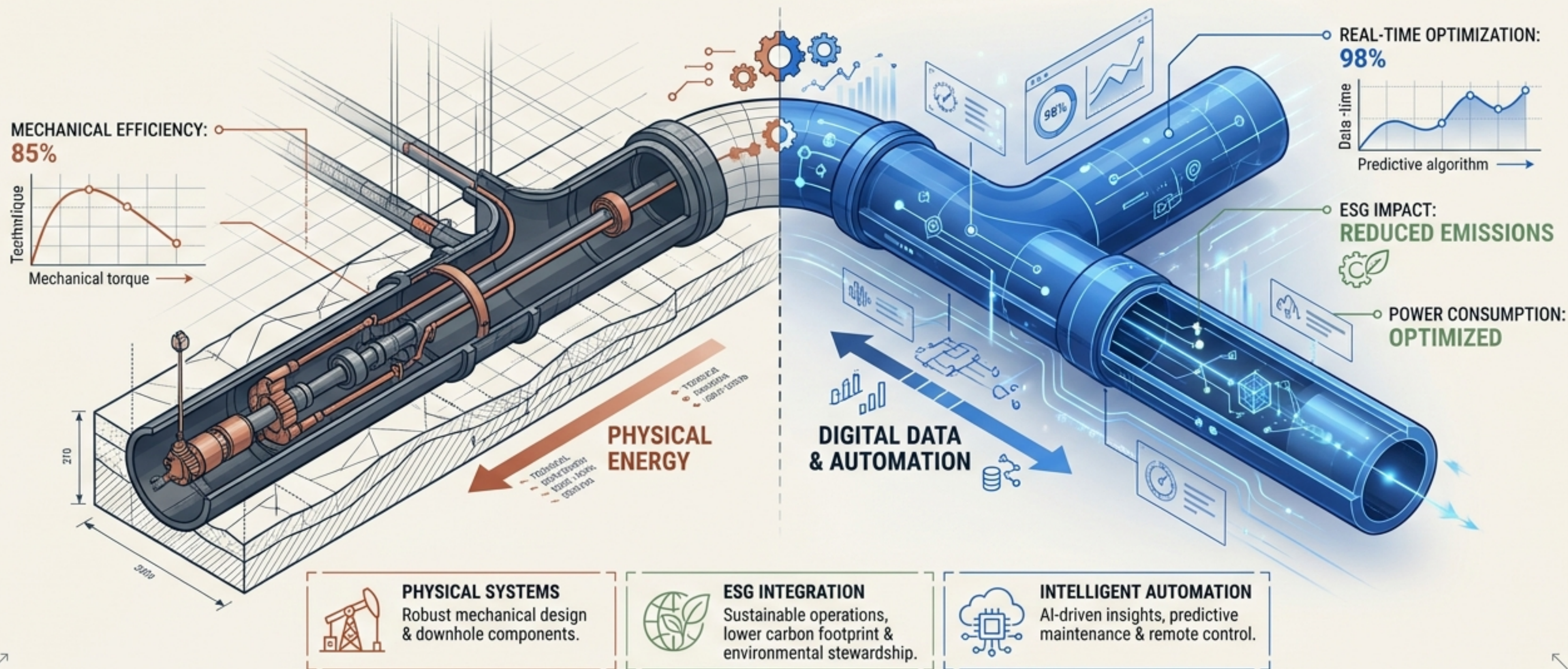


THE CYBER-PHYSICAL WELLBORE

Modernizing Artificial Lift through Data, ESG, and Intelligent Engineering



THE DUAL MANDATE OF THE MODERN OILFIELD

THE RESERVOIR IMPERATIVE

- Overcome decreasing reservoir pressure
- Increase daily production volumes
- Extend overall asset life
- Initiate flow in new wells



THE DIGITAL & ESG IMPERATIVE



Optimize field-wide power consumption



Minimize surface footprint and emissions



Integrate downhole performance with surface SCADA



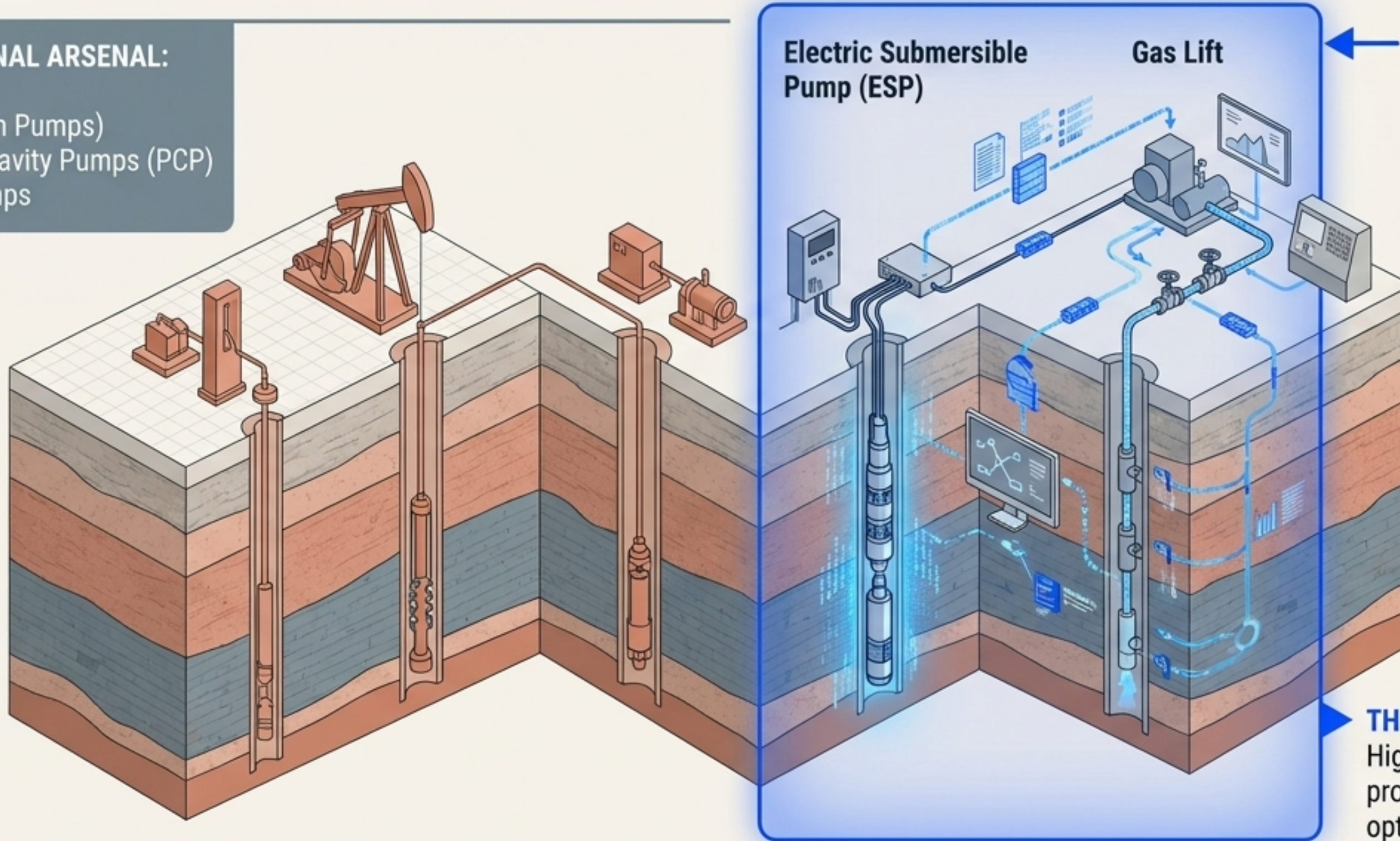
Automate continuous optimization loops



TRANSITIONING FROM MECHANICAL PUMPS TO INTEGRATED CYBER-PHYSICAL NETWORKS

THE TRADITIONAL ARSENAL:

- Plunger Lift
- Rod Lift (Beam Pumps)
- Progressing Cavity Pumps (PCP)
- Hydraulic Pumps



THE FOCUS:

High-volume, automated production architectures optimized for the digital oilfield.

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High-volume, automated production architectures optimized for the digital oilfield.

THE ESP ECOSYSTEM OPERATES ON A CONTINUOUS, AUTOMATED DATA LOOP

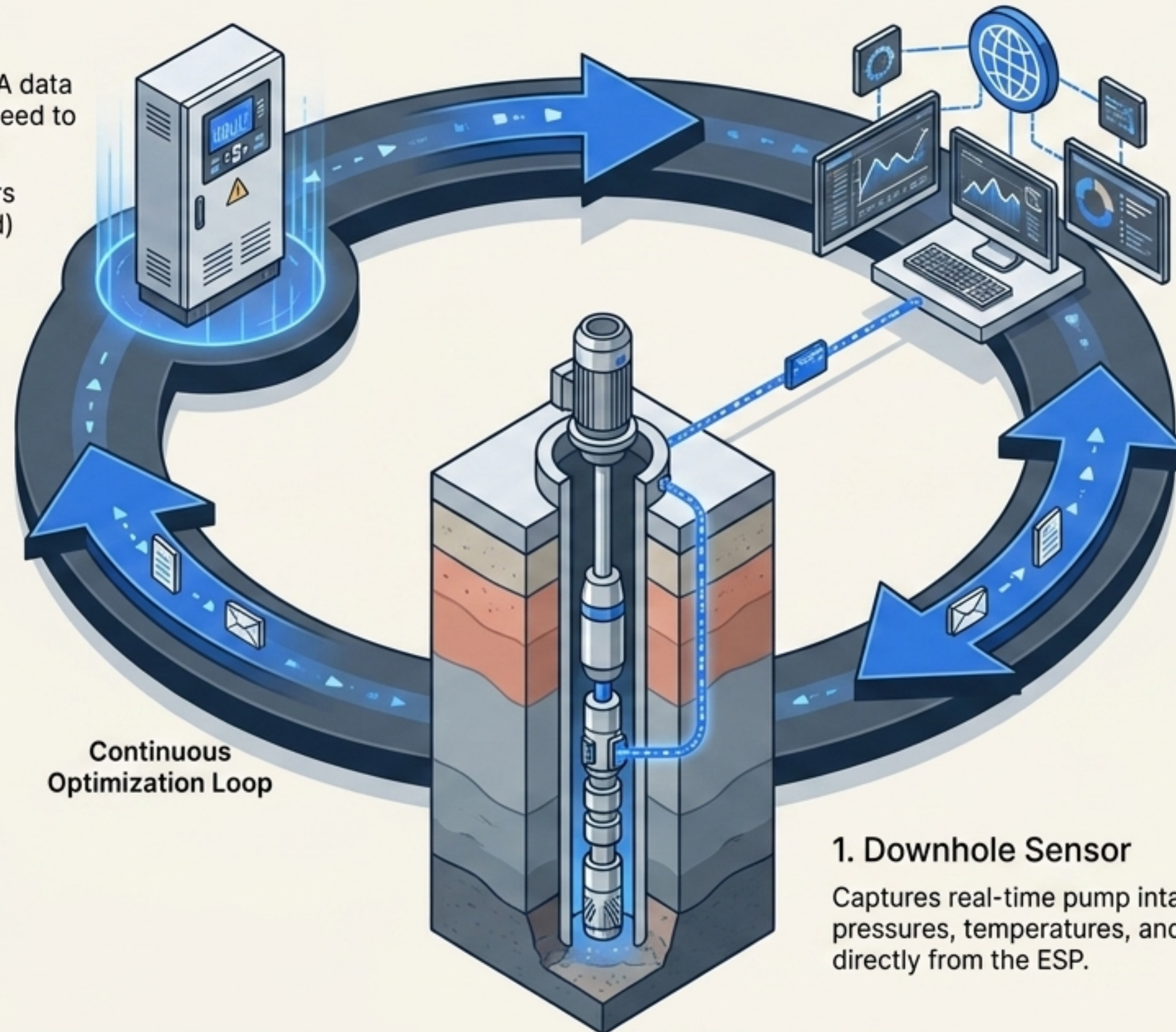
3. Controller Action (VSD)

Variable Speed Drives read SCADA data and automatically adjust motor speed to optimize production continuously.

(Alternatively, soft-start controllers maintain a single protective speed)

2. SCADA Integration

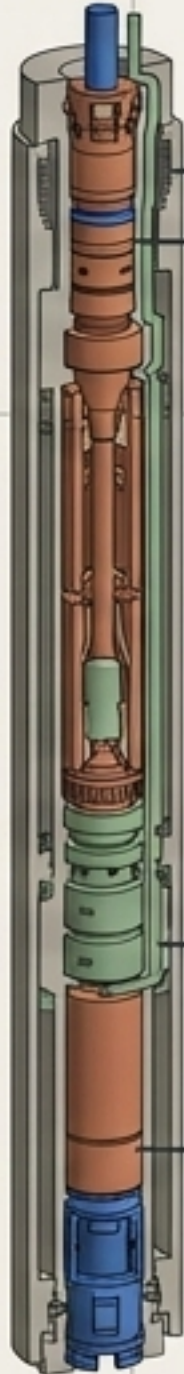
Transmits downhole telemetry securely to surface controllers and central command.



1. Downhole Sensor

Captures real-time pump intake/discharge pressures, temperatures, and vibration directly from the ESP.

DOWNHOLE ARCHITECTURE: ENGINEERING HIGH-VOLUME CENTRIFUGAL LIFT.



Electric Cable

Constructed for harsh environments. Strapped to the production tubing in intervals to prevent mechanical wear.



Check Valve

Threaded into the tubing above the pump; keeps the tubing full of liquid when the pump is idle.

Seal-Chamber

Isolates the motor from well fluids, equalizes internal oil pressure with the wellbore, and absorbs axial thrust.

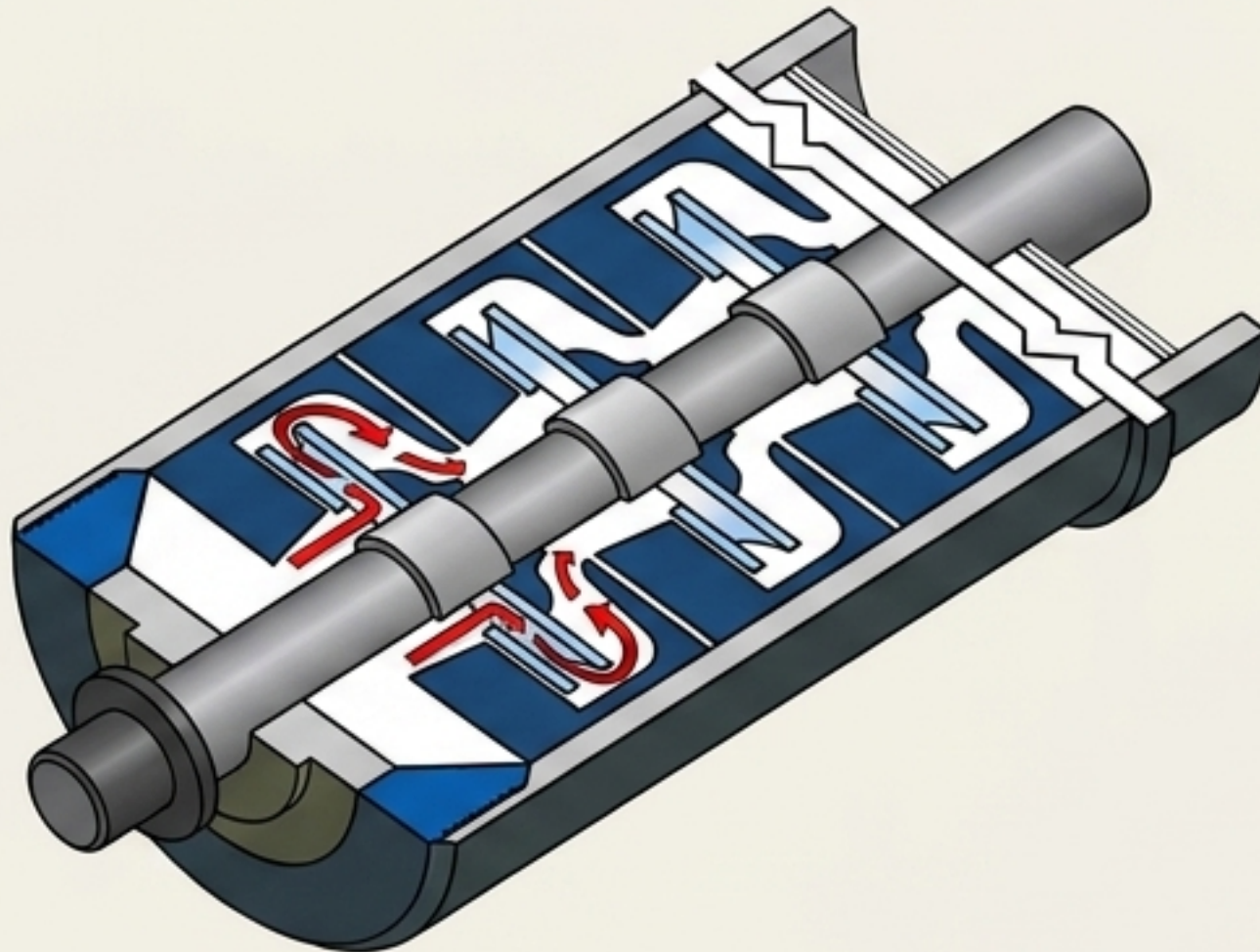
Electric Motor

The power plant. Its horsepower dictates the overall length and diameter of the downhole assembly.

GENERATING TOTAL DEVELOPED HEAD (TDH) THROUGH STAGED KINETIC ENERGY.

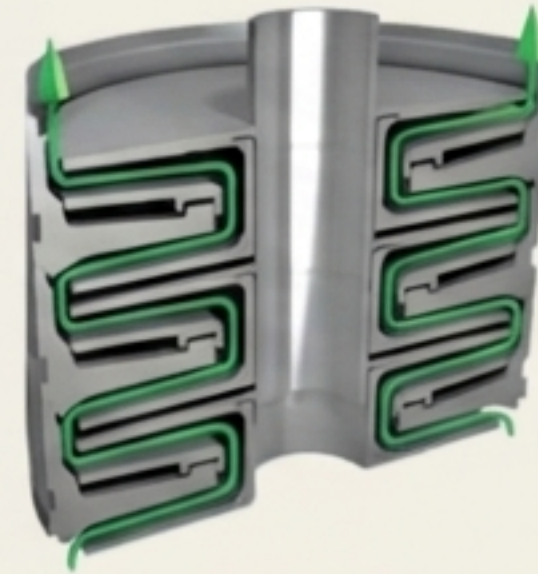
SHAFT MECHANICS

The central shaft connects the motor to the impellers. It is engineered with the smallest possible diameter to maximize fluid bypass volume, without compromising tensile strength.



STAGE ARCHITECTURE

Each stage = Rotating Impeller (increases kinetic energy) + Stationary Diffuser (converts velocity to potential pressure).



RADIAL FLOW

~90-degree vane angles.
Optimized for lower flow rates.

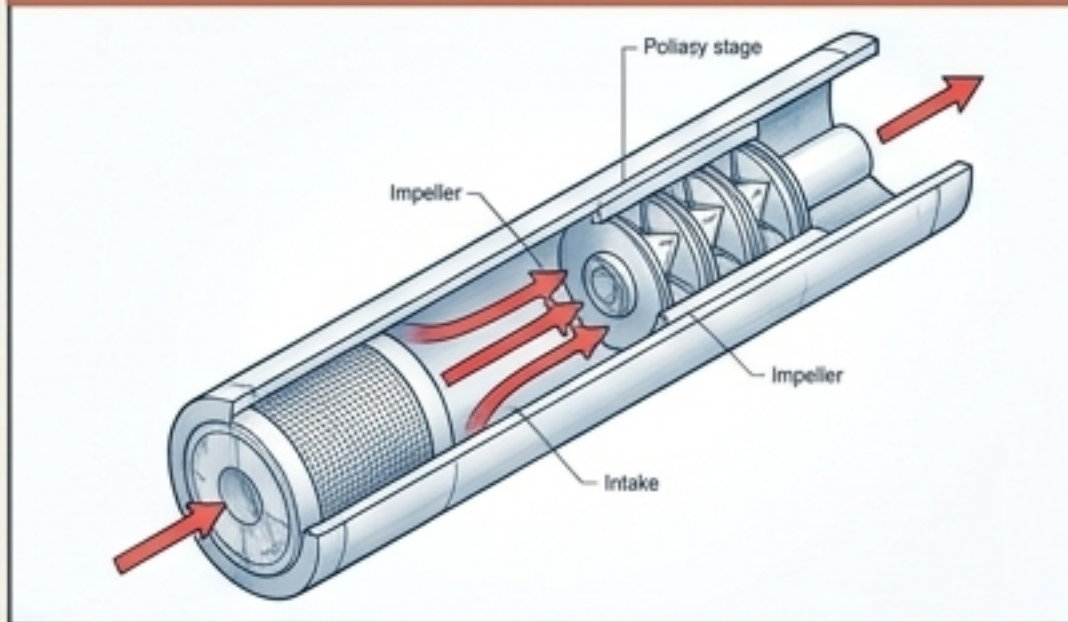


MIXED FLOW

~45-degree vane angles.
Engineered for higher flow rates.

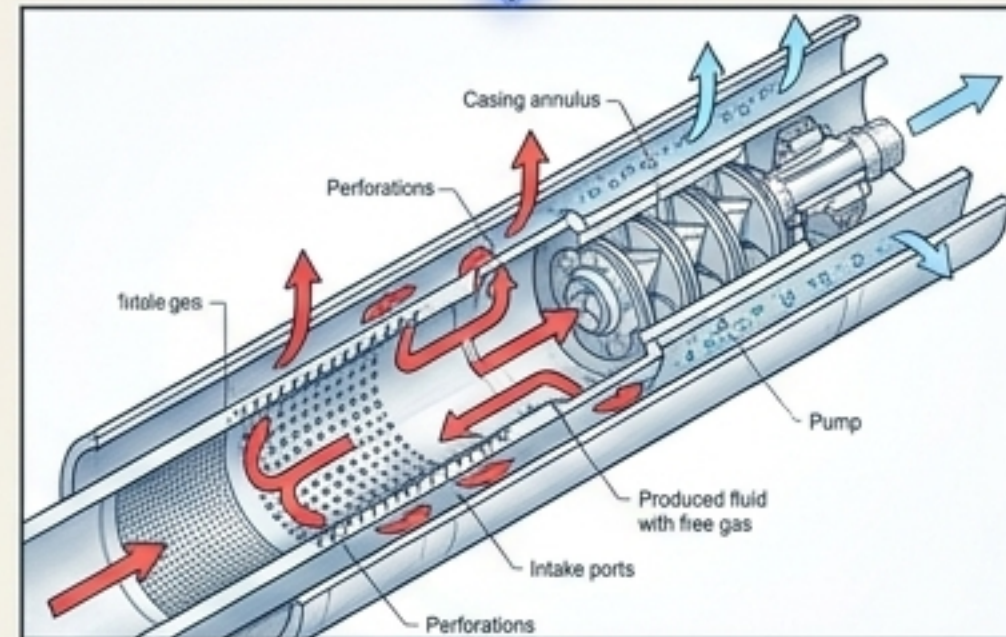
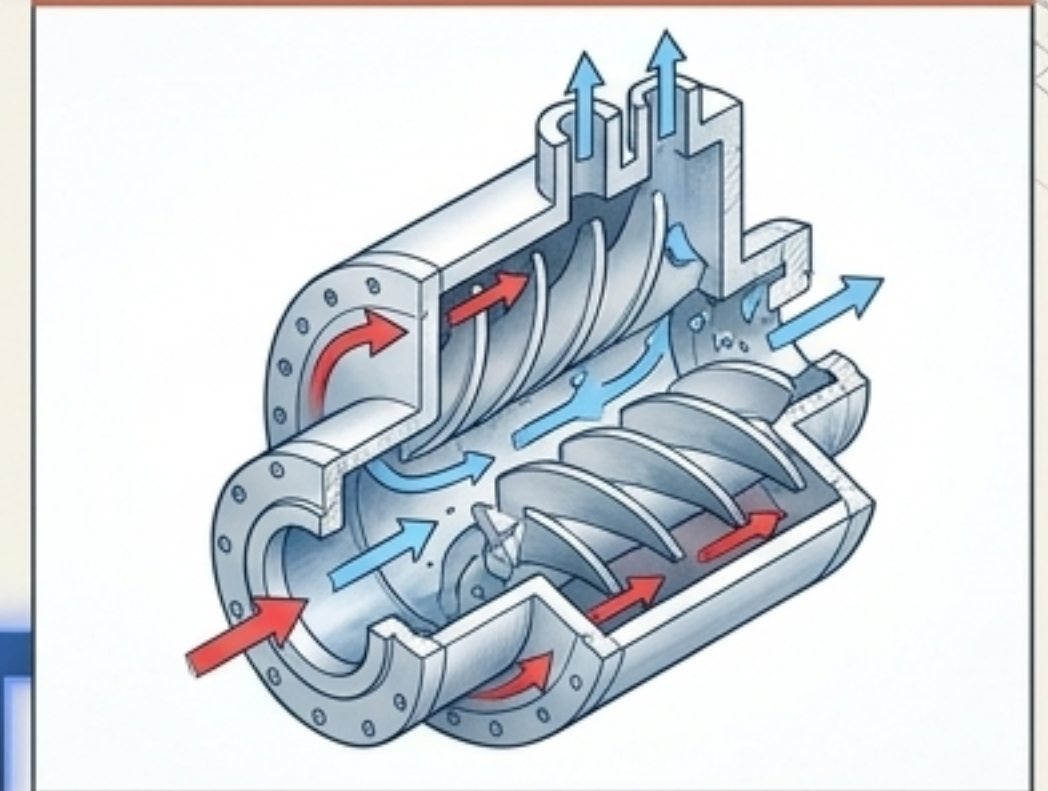
ADAPTING THE INTAKE ARCHITECTURE TO SOLVE THE GAS-LIQUID RATIO (GLR) CHALLENGE

IF **VERY LOW GLR** → Standard Intake



Draws fluid directly into impellers without separating gas.

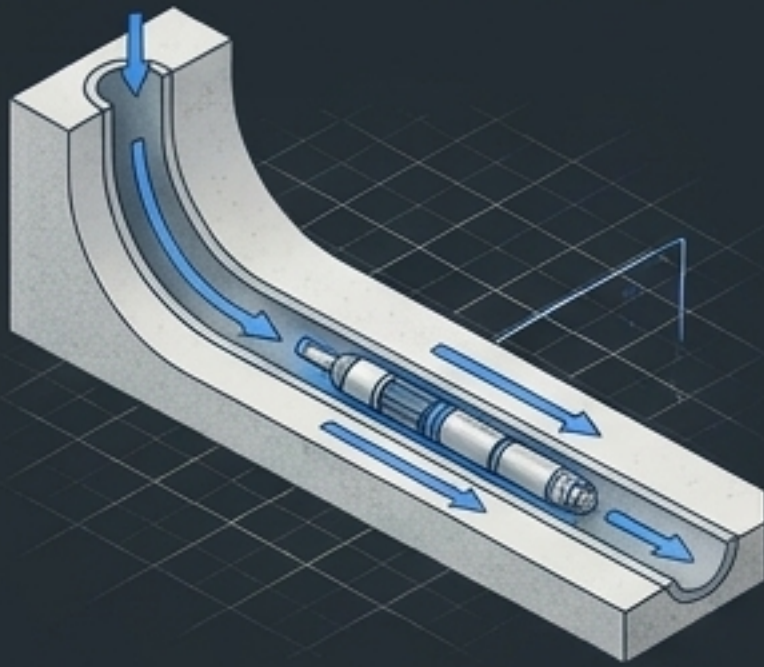
IF **HIGH GLR** → Reverse-Flow Intake



IF **EXTREME GLR** → Rotary Pump Intake (Dynamic Separator)

Operates like a centrifuge. A rotating chamber or induced vortex forces heavier fluid to the outside, while free gas migrates to the center and exits via discharge ports.

THE ESP PROFILE: HIGH VOLUME, LOW FOOTPRINT, AND DYNAMIC ADAPTABILITY

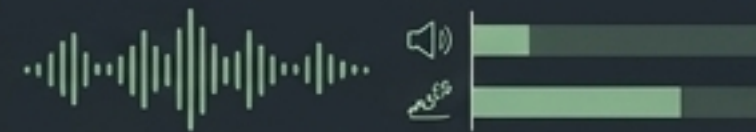


GEOMETRY TOLERANCE

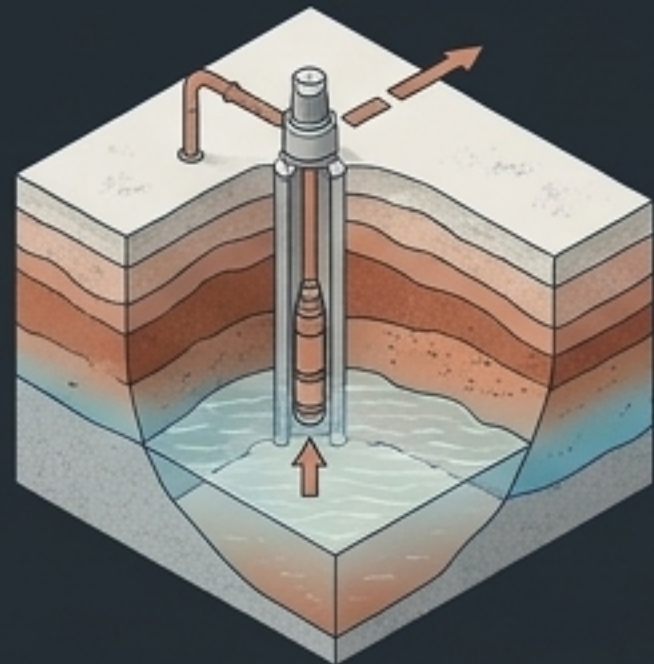


Highly adaptable to severely deviated or horizontal wellbores, provided the actual pump assembly can be set in a relatively straight section of the casing.

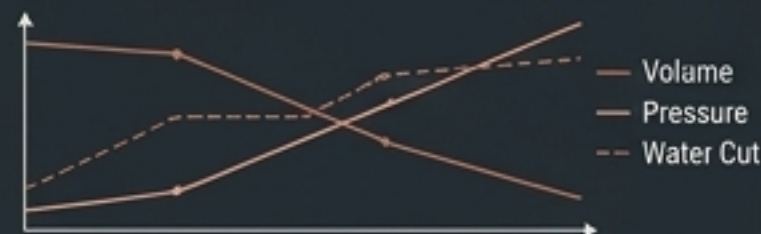
ESG & ENVIRONMENTAL FOOTPRINT



Exceptionally quiet, inherently safe, and requires a minimal surface footprint. The optimal choice for offshore platforms and environmentally sensitive ecosystems.

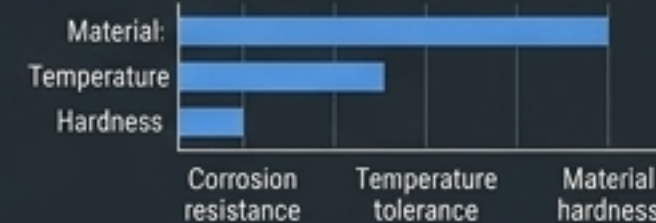


DYNAMIC EVOLUTION

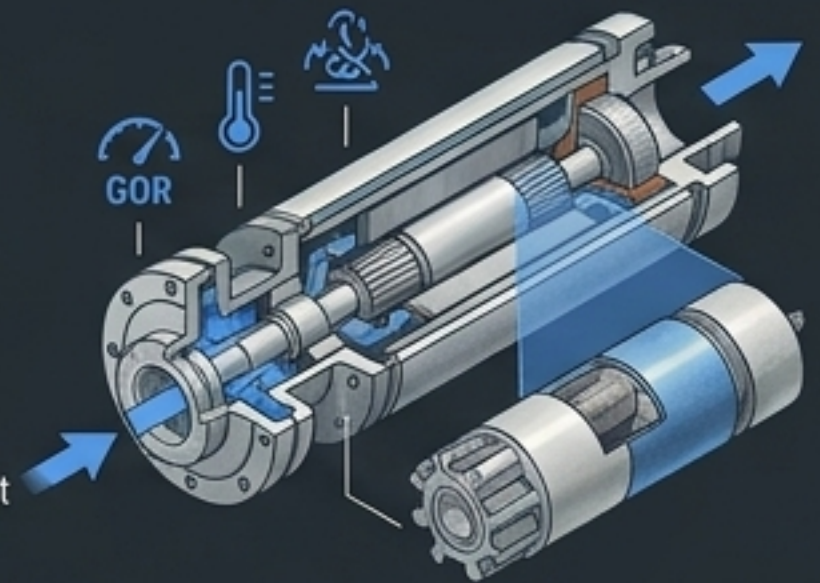


High-volume fluid handling capabilities easily accommodate evolving fluid properties, declining pressures, and increasing water cuts typical during secondary recovery operations.

MATERIAL RESILIENCE



Manufactured from high-grade, corrosion-resistant metallurgical compounds designed specifically to handle high-GOR, extreme downhole temperatures, and highly corrosive acid gases.

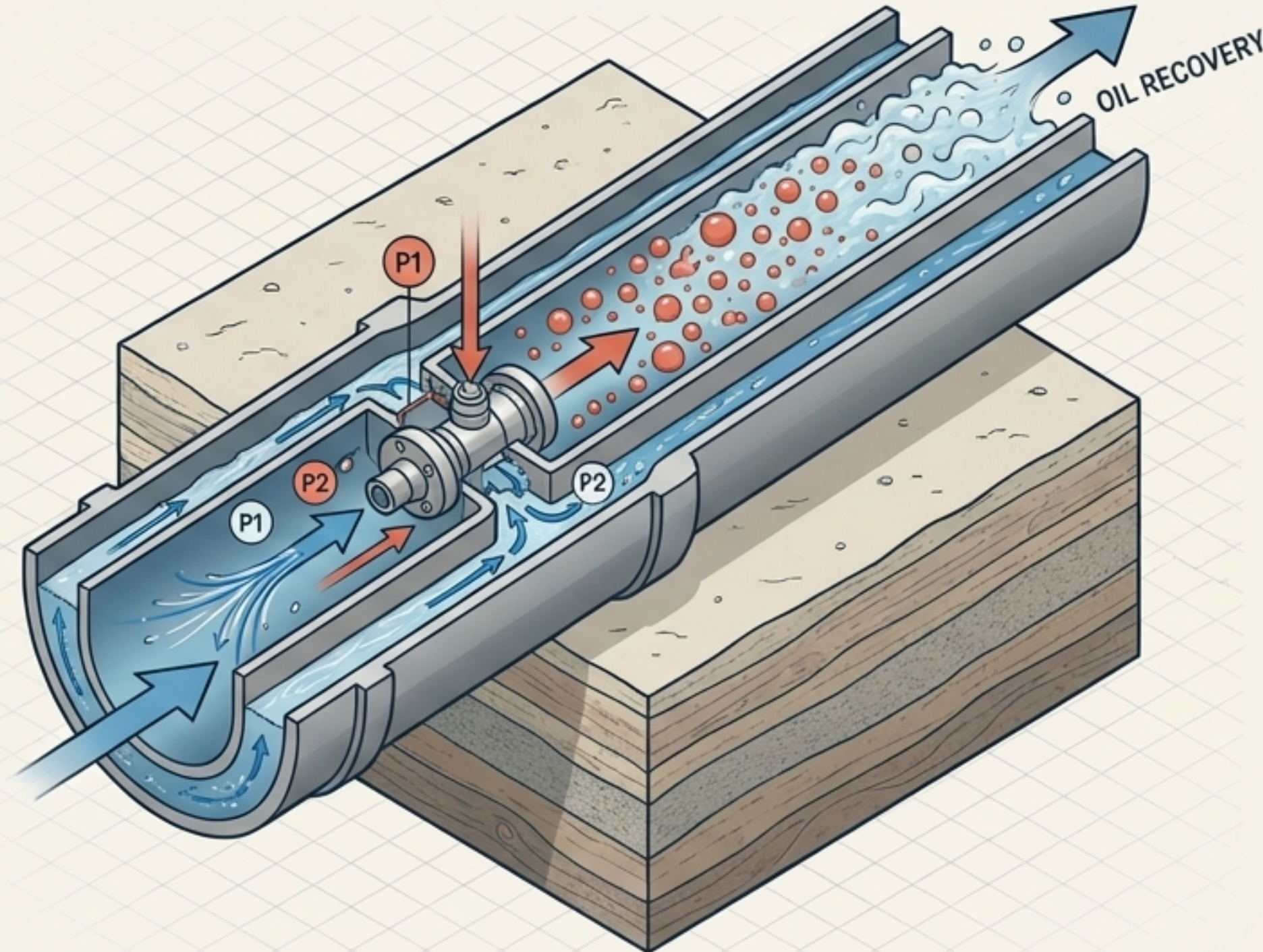


GAS LIFT: THE AUTOMATED PNEUMATIC LUNG OF THE WELLBORE.

THE GOVERNING PHYSICS

Compressed natural gas is injected down the casing annulus and enters the production tubing through specific valves.

The expanding gas propels oil upward while actively aerating the fluid column—drastically lowering its density so it flows to the surface with less required reservoir pressure.



OPERATIONAL MODES

1. CONTINUOUS FLOW

Ideal for high GLR, highly inclined wells, and handles sand or scale effortlessly. (Not suitable for heavy or emulsion oils).

2. INTERMITTENT FLOW

Designed for low-pressure, low-productivity formations. Gas is injected in automated, predetermined cycles.

MANAGING THE CLOSED-LOOP PNEUMATIC ARCHITECTURE AT THE SURFACE.

MAKEUP GAS SOURCING

If the well produces insufficient volume, supplementary gas must be routed from commercial sales lines or neighboring well pipelines simply to kickoff the process.

Sales Line
Neighboring Well or Pipeline

MAKEUP GAS SOURCING

If the well produces insufficient volume, supplementary gas must be routed from commercial sales lines or neighboring well pipeline simply to kickoff the process.

Gas Injection Compressor

THE COMPRESSOR

Pressurizes low-pressure natural gas for the injection line. This is a high-CAPEX, incredibly energy-intensive component that heavily impacts the site's overall ESG footprint.

Gas Injection Control Valve

GAS INJECTION CONTROL VALVE

The digital gateway. Automates and precisely throttles the exact flow of compressed natural gas into the well casing.

Separator

Gas Lift Valve

Packer

Standing Valve

FORMATION

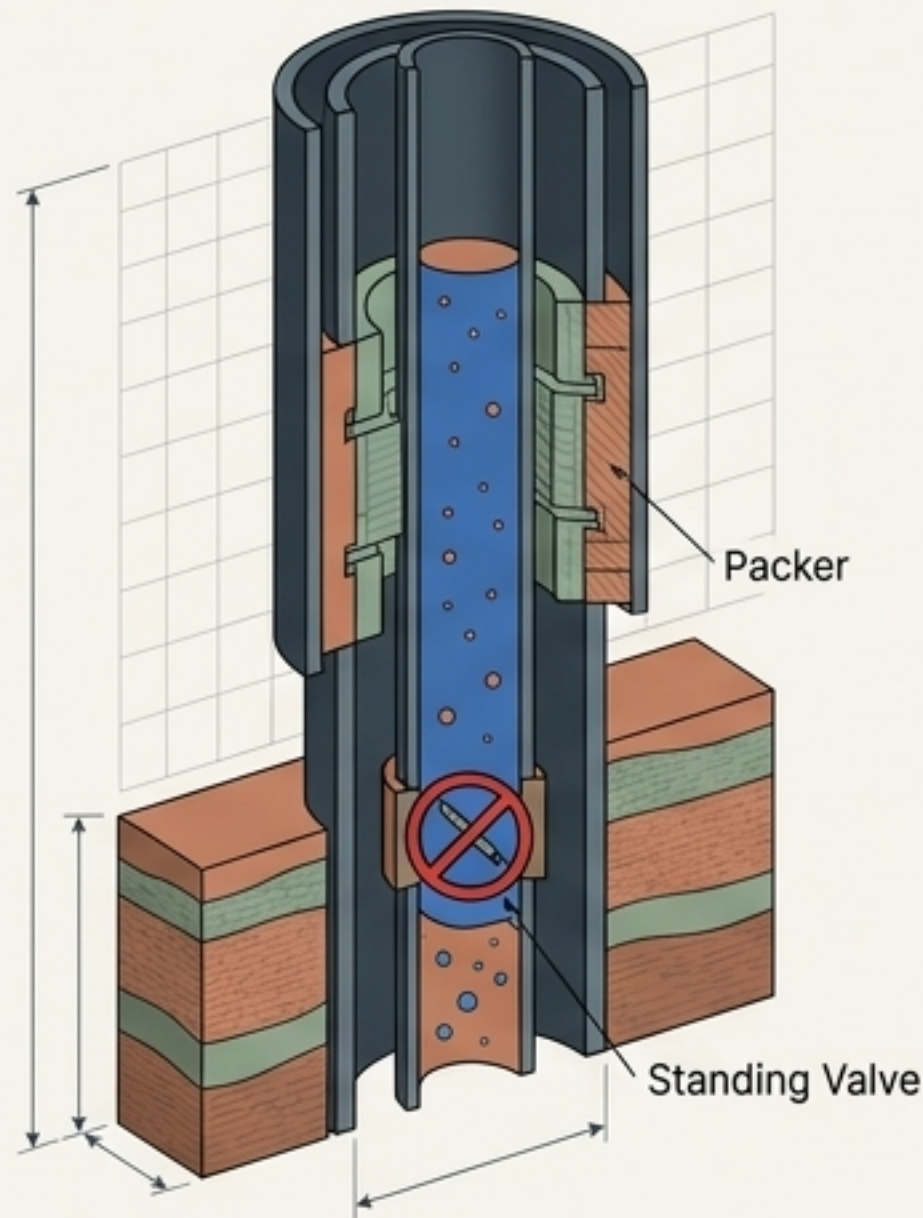
Gas Injection

Natural Gas Injection Line

WELL KICKOFF: THE DYNAMIC UNLOADING SEQUENCE

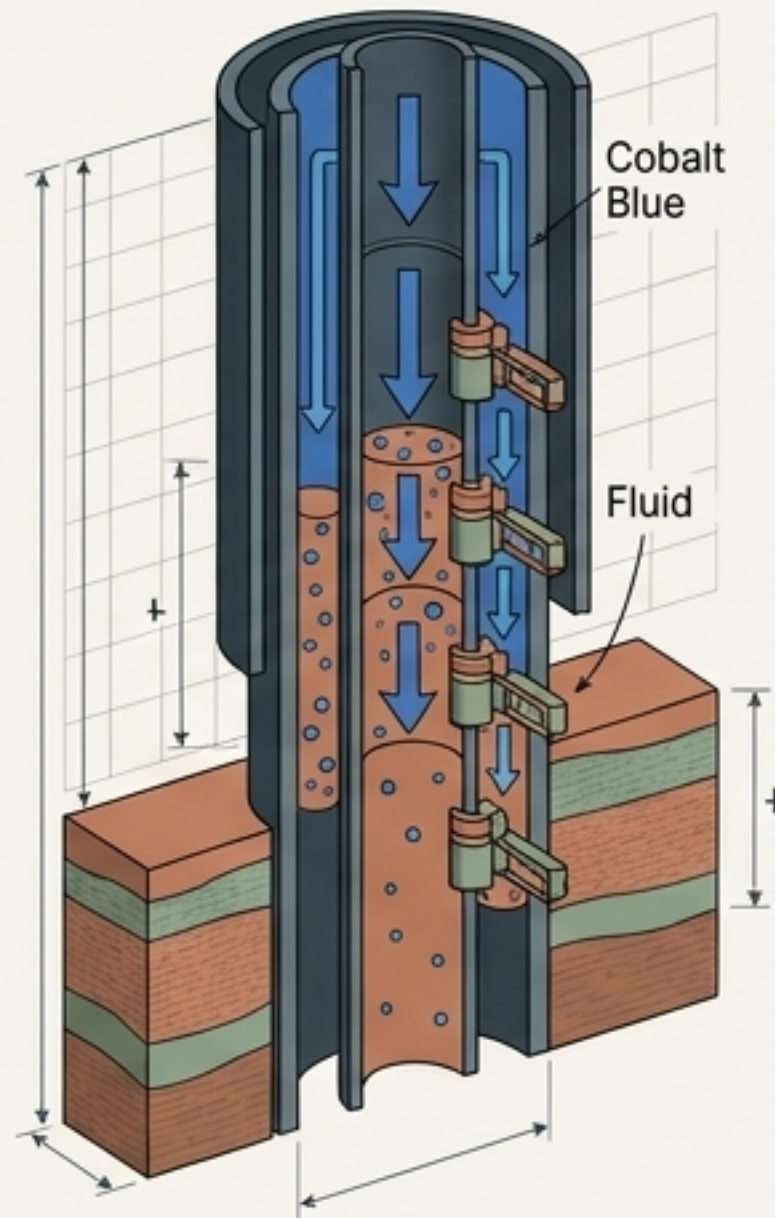
The dynamic unloading sequence.

1. COMPLETE ISOLATION



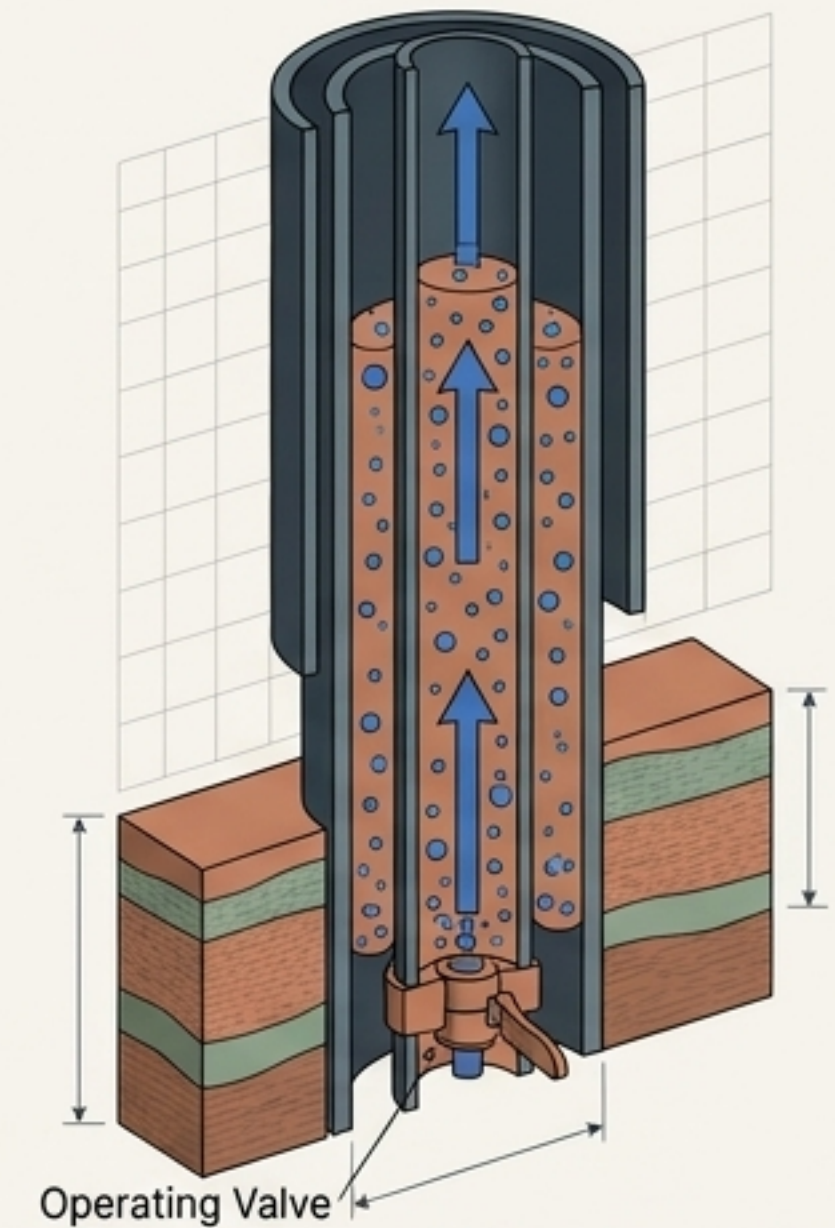
A **Packer** forms a permanent seal between the casing and tubing. The **Standing Valve** prevents fluid from exiting back into the formation, keeping the system pressurized.

2. SEQUENTIAL UNLOADING



Injected gas moves downward, forcing sequential unloading valves to open. This systematically aerates and reduces the density of the fluid column staged above.

3. OPERATING STATE



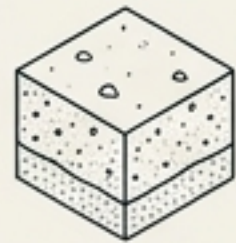
Gas reaches the deepest valve (the **Operating Valve**). This achieves continuous well kickoff, injecting gas at the lowest possible point to maximize lift.

THE GAS LIFT PROFILE: HIGH RELIABILITY AGAINST HIGH INFRASTRUCTURE DEMANDS.

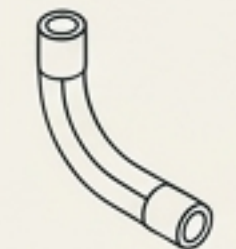
STRATEGIC ADVANTAGES



- **High Reliability:** Fewer moving parts downhole minimizes catastrophic failure risk.



- **Material Tolerance:** Uniquely tolerant to sand, debris, and extreme downhole temperatures.

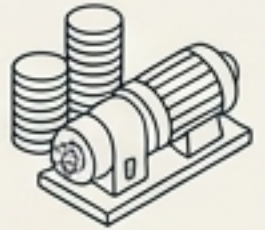


- **Geometry Agnostic:** Excels in highly deviated and horizontal wellbores without mechanical wear.



OPERATIONAL CONSTRAINTS

- **High CAPEX:** Massive initial capital required for compressors and surface infrastructure.



- **Energy Intensive:** Continuous compression significantly increases operating costs and carbon emissions.



- **Large Footprint:** Unsuitable for compact offshore platforms due to spatial requirements.



- **Supply Dependency:** Requires absolute reliability of makeup gas availability.



THE LIFT SELECTION MATRIX: ALIGNING TECHNOLOGY WITH WELLBORE REALITY.

EVALUATION METRIC	ELECTRIC SUBMERSIBLE PUMP (ESP)	GAS LIFT
Space & ESG Footprint	Minimal footprint, practically silent. Extremely low emissions.	Massive footprint. High emissions due to heavy continuous compression.
Well Geometry Tolerance	Adaptable, but absolutely requires a straight section of casing to set the rigid assembly.	Highly adaptable. Excels in severe deviation and complex horizontal trajectories.
Solids & Sand Handling	Highly sensitive to erosion. Requires specialized, expensive shaft and pump metallurgies.	Extremely tolerant. Easily handles high volumes of sand, wax, and scale.
CAPEX vs OPEX Profile	Lower initial surface CAPEX. Moderate OPEX driven by grid electricity.	Extremely high initial CAPEX (infrastructure). High OPEX driven by continuous fuel consumption.

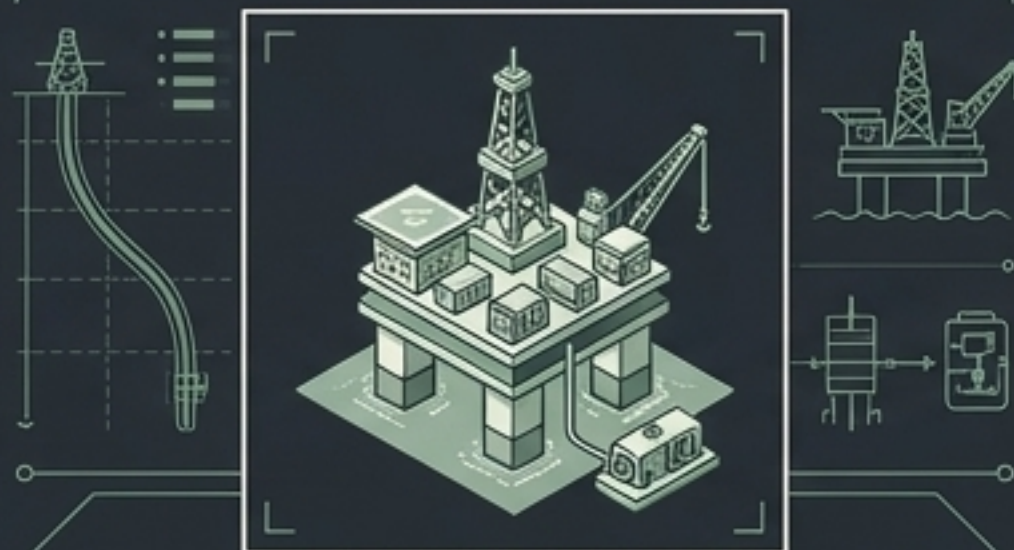
MULTIDISCIPLINARY ALIGNMENT: THE 3-EXPERT PANEL



THE RESERVOIR ENGINEER

KPI Focus: Maximum Ultimate Recovery

"Gas lift is brilliant for our high-GLR and sandy wells, but as pressure depletes and water cuts rise, the ESP's Total Developed Head is non-negotiable for maximizing our secondary recovery."



THE PRODUCTION ENGINEER

KPI Focus: Geometry, Uptime & Space

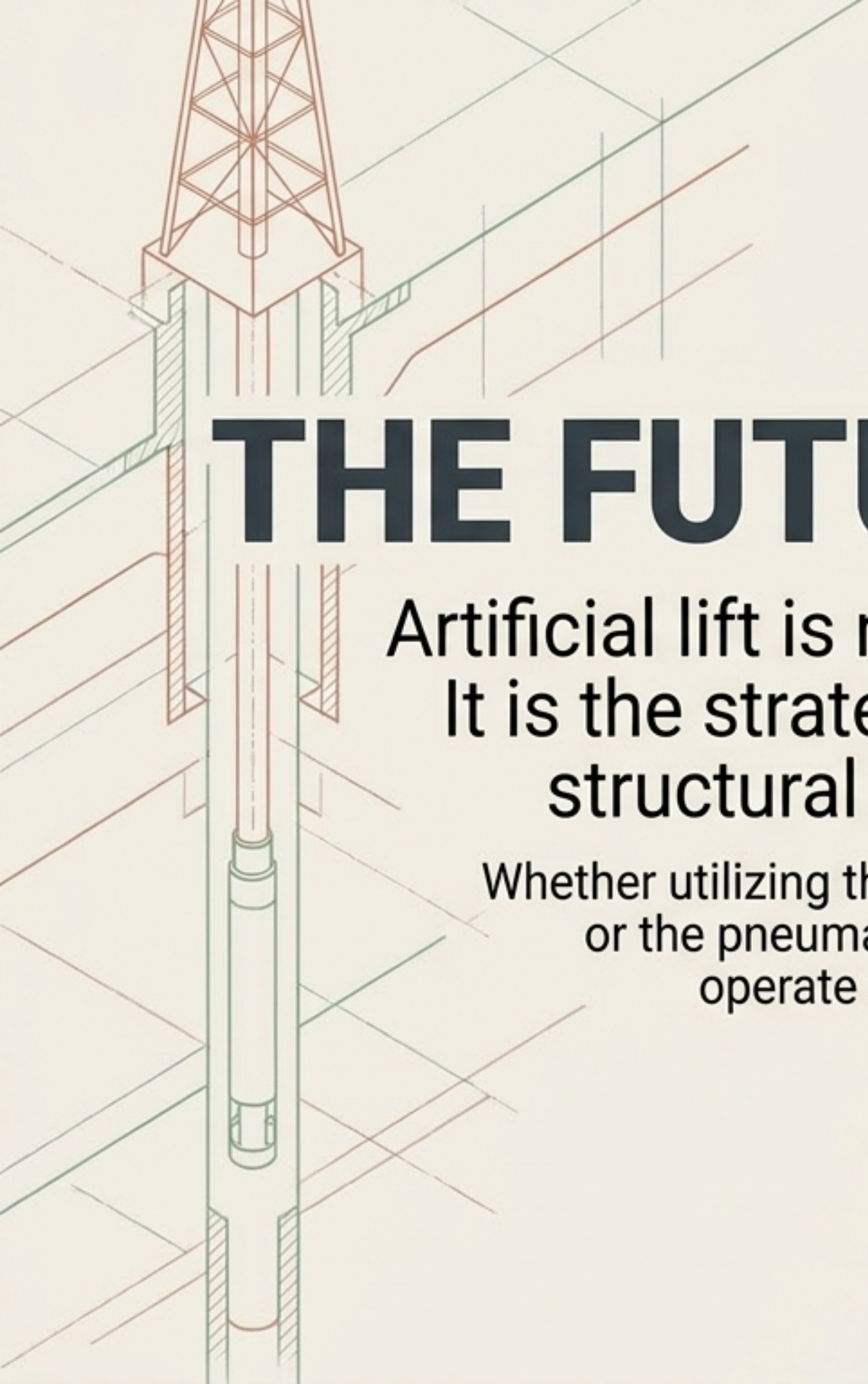
"Our highly deviated offshore wells have zero surface space for massive compressors. The ESP's small footprint solves our spatial constraints—provided we can find a straight section to set it."



THE AUTOMATION ENGINEER

KPI Focus: Cyber-Physical Control

"Both systems are now digital nodes. Gas lift relies on automated surface injection, but an ESP with downhole sensors feeding a SCADA-linked VSD offers a much tighter, faster closed-loop optimization."



THE FUTURE-READY ASSET

Artificial lift is no longer a purely mechanical choice.
It is the strategic integration of reservoir physics,
structural geometry, and automation logic.

Whether utilizing the data-driven engine of an Electric Submersible Pump
or the pneumatic adaptability of Gas Lift, the modern well must
operate as an intelligent, ESG-compliant ecosystem.